11.28.

a. Demand: $Q_i = \alpha_1 + \alpha_2 P_i + \alpha_3 PS + \alpha_4 DI + e_d$

$$\Rightarrow \alpha_2 P = Q - \alpha_1 - \alpha_3 PS - \alpha_4 DI - e_d$$

$$\Rightarrow P = \frac{Q - \alpha_1 - \alpha_3 PS - \alpha_4 DI - e_d}{\alpha_2}$$

$$\Rightarrow P = a_1 + a_2 Q + a_3 PS + a_4 DI + \varepsilon_d$$

$$a_2 = \frac{-\alpha_1}{\alpha_2} < 0 \,,\ a_3 = \frac{1}{\alpha_2} > 0 \,,\ a_4 = \frac{-\alpha_4}{\alpha_2} > 0$$

Supply: $Q_i = \beta_1 + \beta_2 P + \beta_3 PF + e_s$

$$\Rightarrow \beta_2 P = Q_i - \beta_1 - \beta_3 PF - e_s$$

$$\Rightarrow P = \frac{Q_i - \beta_1 - \beta_3 PF - e_s}{\beta_2}$$

$$\Rightarrow P = b_1 + b_2 Q + b_3 PF + \varepsilon_s$$

$$b_2 = \frac{1}{\beta_2} > 0 \;;\ b_3 = \frac{-\beta_3}{\beta_2} > 0$$

#